

SPARC-calibrated galactic wormholes in Einstein–Cartan gravity with Dekel–Zhao and Hernquist dark-matter halos

Yohenba Soibam ^a , M. Zeeshan Gul ^{b c d} , Kangujam Priyokumar Singh ^a   [Show more](#)  Cite  Add to Mendeley  Share | 10.1016/j.dark.2026.102461 

Access through your organi...

Article preview

Abstract

We construct static, spherically symmetric galactic wormholes in Einstein–Cartan gravity using dark-matter halo distributions calibrated from SPARC rotation curves. The Dekel–Zhao (DZ) and Hernquist profiles are fitted to NGC3198 and DDO154 through Bayesian posterior sampling. Hernquist is mildly favored for NGC3198, whereas DZ is preferred for DDO154. Joint posterior samples preserving parameter covariance are propagated into the Einstein–Cartan reconstruction, with both halo families used to assess model dependence. The throat radius is parameterized by $q_0 = r_0/R_{\min}^{SPARC}$, while the matching radius is fixed at $R_{\max}^{SPARC}$. Imposing the smooth Levi–Civita metric-sector condition $p_{r,eff}(R_m) = 0$ determines the Weyssenhoff throat-spin amplitude. Across $0.1 \leq q_0 \leq 1$, all propagated realizations satisfy the adopted geometrical admissibility and positive-mass conditions. At $q_0 = 0.5$, the material and effective radial null energy conditions are violated at the throat. The material WEC and SEC are recovered by the matching radius, whereas the material DEC and all effective WEC, SEC, and DEC remain violated. The exterior masses are approximately $1.75 \times 10^{11} M_\odot$ for NGC3198 and $3.04 \times 10^9 M_\odot$ for DDO154. Changing the halo family modifies these masses by about 4.0% and 0.34%, respectively, without altering the qualitative throat or optical conclusions. The throat supports a radially unstable circular null orbit, with no additional photon sphere in the retained interior or Schwarzschild exterior. A near-throat expansion yields a logarithmic strong-deflection law with $\bar{a}_{th} \simeq \sqrt{2}$. These results constitute a SPARC-calibrated, model-conditioned Einstein–Cartan reconstruction rather than direct observational evidence for wormholes or microscopic spin.

Keywords

Article preview

[Abstract](#)[Introduction](#)[Section snippets](#)[References \(63\)](#)

Introduction

Traversable wormholes are nontrivial spacetime geometries characterized by a minimum areal radius connecting two spacetime regions. In the Morris–Thorne construction, regular traversal requires a finite redshift function together with a shape function satisfying the flare-out condition at the throat [1], [2]. Within general relativity, this geometrical requirement implies violation of the radial null energy condition at or near the throat [3], [4], [5]. Modified gravitational dynamics can alter the relation between the material stress tensor and the effective source of the geometry, making it important to distinguish material energy conditions from their effective counterparts. Finite-radius matching to an exterior vacuum spacetime also provides a natural way to localize non-asymptotically flat wormhole interiors [6].

Einstein–Cartan gravity offers a particularly natural framework in which this distinction can be investigated. Cartan generalized the Riemannian formulation of gravitation by allowing the affine connection to possess an antisymmetric part, identified with spacetime torsion [7]. The gauge-theoretic interpretation of gravity developed by Kibble and Sciama established a direct relation between intrinsic angular momentum and torsion [8], [9]. In the resulting Einstein–Cartan theory, the metric is sourced by energy and momentum, whereas torsion is sourced algebraically by the spin density of matter. Consequently, torsion does not propagate independently in vacuum in the minimal theory, but generates local corrections quadratic in spin after it is eliminated from the field equations [10], [11], [12]. A convenient macroscopic description of matter carrying intrinsic spin is provided by the Weyssenhoff fluid, originally formulated by Weyssenhoff and Raabe and subsequently developed within a systematic variational Einstein–Cartan framework [13], [14]. Covariant treatments of this fluid show that its averaged spin contribution can modify the effective density and pressures while preserving a fluid description at macroscopic scales [15].

The possible role of spin and torsion in sustaining wormhole geometries has consequently received increasing attention. Einstein–Cartan solutions supported without conventional phantom fields have been obtained for nonminimally coupled scalar and electromagnetic sectors [16]. Static wormholes sourced by a Weyssenhoff fluid have been shown to admit parameter regions in which the ordinary material variables satisfy the standard energy conditions even though the effective source governing the throat violates the corresponding effective condition [17]. Dynamical extensions have also been constructed in which spin corrections influence the temporal evolution of the throat [18]. More recently, Yukawa–Casimir matter distributions have been considered in Einstein–Cartan gravity together with energy-condition, equilibrium, embedding, tidal, and optical diagnostics [19]. Related compact-object studies in alternative gravitational frameworks have also shown that departures from general relativity can modify orbital dynamics and quasi-periodic-oscillation phenomenology, including in Horndeski and Hu–Sawicki gravity [20], [21].

Galactic rotation curves provide one of the principal observational probes of the dark-matter distribution [22], [23], [24]. In particular, the SPARC database combines spatially resolved rotation curves with homogeneous near-infrared photometry for a large sample of disk galaxies and therefore provides a useful basis for constraining halo models [25]. A wide variety of spherical halo profiles has been developed to describe galactic dark-matter distributions, including the Einasto, NFW, Burkert, and more general double-power-law families [26], [27], [28], [29], [30], [31]. In the present work we consider two spherical profiles with complementary properties. The Dekel–Zhao (DZ) profile provides a flexible description of both cuspy and cored inner halo structures while retaining analytical expressions for several dynamical quantities [32]. The Hernquist profile, by contrast, is a finite-mass spherical model with simple closed-form density, enclosed-mass, and potential functions [33], [34]. These properties make the two profiles particularly convenient for propagating rotation-curve constraints into an Einstein–Cartan wormhole reconstruction.

Dark-matter profiles have already been employed as candidate sources for galactic wormholes. Wormhole geometries compatible with halo regions and approximately flat galactic rotation curves were studied in general relativity by Rahaman and collaborators [35], [36], and their lensing properties were considered by Kuhfittig [37]. Bose–Einstein-condensate and other halo models have also been examined as possible wormhole sources [38], [39]. More recently, the Dekel–Zhao profile has been used to construct wormholes in general relativity, generalized geometry–matter coupling, symmetric teleparallel gravity, and quantum-improved gravity [40], [41], [42], [43]. The Hernquist profile has likewise been proposed as a source for galactic wormhole configurations [44]. Photon dynamics and shadow morphology have additionally been explored for slowly rotating wormholes immersed in cuspy and cored dark-matter environments [45]. Complementary compact-object studies have also examined how surrounding dark-matter distributions affect particle dynamics and oscillatory observables, including Schwarzschild-type geometries embedded in Dehnen-type and perfect-fluid dark-matter environments [46], [47]. Despite this progress, most wormhole studies select representative halo parameters directly; the uncertainties of observationally fitted rotation-curve parameters are rarely propagated into the reconstructed wormhole geometry.

The optical behavior of wormholes provides an additional means of characterizing their geometry. Light propagation in the Ellis wormhole was investigated in early studies of geometrical optics [48], and subsequent analyses derived weak- and strong-field deflection angles for static wormholes [49], [50]. Lensing in scalar–tensor wormholes has also illustrated how the matter sector and the energy conditions affect observable photon trajectories [51]. More generally, the logarithmic strong-deflection formalism developed for static spherical spacetimes provides a systematic method for extracting critical impact parameters and relativistic-image observables [52]. Strong-lensing studies have exhibited critical-curve structures in scalar-field spacetimes [53], while the geometric theory of photon surfaces clarifies the role of circular null trajectories in static spacetimes [54]. Related compact-object studies have examined photon motion and shadows of regular black holes, weak gravitational lensing in quantum-corrected geometries, and plasma lensing around short-hairy black holes [55], [56], [57]. For a galactic redshift potential inferred from a nonrelativistic rotation curve, an ordinary off-throat photon-sphere condition is not generally expected to be satisfied. The relevant critical orbit may instead be

supported by the wormhole throat itself, making a proper-radial-distance analysis essential.

Motivated by these considerations, the present work develops a SPARC-calibrated, model-conditioned reconstruction of static galactic wormholes in Einstein–Cartan gravity. We fit both the Dekel–Zhao and Hernquist halo models to the rotation curves of NGC3198 and DDO154 and propagate joint posterior samples, preserving their parameter covariance, into the wormhole sector. For the primary reconstruction we adopt the mildly preferred Hernquist posterior for NGC3198 and the DZ posterior for DDO154 marginalized over an intrinsic velocity-scatter parameter. The alternative halo family is also propagated for each galaxy as a discrete model-dependence test.

Following earlier phenomenological constructions in which galactic dark-matter profiles are employed as matter sources for wormhole geometries [35], [36], [39], the spherical halo density is employed here as a phenomenological material source for the Weyssenhoff sector. The halo circular velocity determines the spherical contribution to the redshift potential through the circular-geodesic relation derived in Section 3.2. These identifications, together with the spin-profile closure introduced below and the phenomenological throat scale $q_0 = r_0/R_{\min}^{SPARC}$, constitute explicit modeling assumptions rather than quantities directly measured by SPARC.

Finite-radius localization of a wormhole interior by matching to an exterior vacuum geometry has standard precedent [6]. In the present construction, however, the specific choice $R_m = R_{\max}^{SPARC}$ is an observationally motivated prescription introduced to avoid extrapolating the fitted halo beyond the last measured rotation-curve point. The dimensionless throat-spin amplitude is then determined by smooth matching of the Levi–Civita metric sector to a Schwarzschild exterior.

The complete halo-posterior uncertainty is propagated into the shape function, flare-out condition, material and effective energy conditions, exterior mass, critical impact parameter, and local strong-deflection coefficient. We additionally quantify the sensitivity of the redshift potential to the omitted baryonic contribution and test the dependence of the Einstein–Cartan reconstruction on the discrete choice of halo profile. Finally, the throat-supported circular null orbit and its strong-deflection limit are analyzed analytically and numerically.

The paper is organized as follows. Section 2 summarizes the Einstein–Cartan field equations, the Weyssenhoff spin sector, and the static wormhole geometry. Section 3 describes the SPARC data, halo models, Bayesian likelihood, posterior constraints, and halo-derived redshift potentials. Section 4 develops the posterior-matched shape function, geometrical admissibility conditions, material and effective energy conditions, and the smooth metric-sector Schwarzschild junction. The throat light ring and strong gravitational deflection are studied in Section 5. The numerical results and their physical interpretation are presented in Sections 6, and 7 summarizes the main conclusions.

Organizational access

Get full-text access by signing in with your organisation

Section snippets

Spin–torsion sector and field equations

Einstein–Cartan gravity extends general relativity from a Riemannian spacetime to a Riemann–Cartan spacetime in which the affine connection is not assumed to be symmetric. Following the standard Einstein–Cartan formulation [11], [12], we consider the action

$$\mathcal{S} = \int d^4x \sqrt{-g} \left[-\frac{\widetilde{R}}{2\kappa} + \mathcal{L}_m \right],$$

where $\kappa = 8\pi G_N/c^4$, $\widetilde{R}$ is the Ricci scalar constructed from the full affine connection, and $\mathcal{L}_m$ is the matter Lagrangian. We set the cosmological constant to zero as a modeling choice so that the vacuum exterior used below is ...

Halo models, data, and likelihood

The observational calibration is based on the public galaxy-by-galaxy SPARC rotation-curve data [25]. The present baseline analysis holds the published galaxy distances and inclinations fixed. We consider NGC3198 as a representative extended spiral galaxy and DDO154 as a representative dark-matter-dominated dwarf galaxy. Neither galaxy has a bulge contribution in the adopted SPARC rotation-curve files.

The dark-matter distribution is modeled using either the Dekel–Zhao or Hernquist profile. The ...

Posterior-matched shape function and admissibility

The SPARC analysis determines the halo density and redshift potential but does not fix the wormhole throat scale. We parameterize the throat radius by

$$q_0 = \frac{r_0}{R_{\min}^{\text{SPARC}}}, \quad 0.1 \leq q_0 \leq 1,$$

so that

$$r_0 = q_0 R_{\min}^{\text{SPARC}}.$$

The interval $0.1 \leq q_0 \leq 1$ is a phenomenological throat-scale prior introduced to keep the throat at or below the innermost rotation-curve measurement while limiting the extent of inward extrapolation. It is not inferred from a SPARC observable. For NGC3198 and DDO154 we have $R_{\min}^{\text{SPARC}} = 0.32 \text{ kpc}$ and ...

Throat light ring and critical impact parameter

We consider equatorial null geodesics in the static wormhole geometry of Eq. (18).

The conserved energy and angular momentum per unit affine parameter are

$$E = e^{2\Phi(r)} \dot{t}, \quad L = r^2 \dot{\phi}, \quad u = \frac{L}{E},$$

where an overdot denotes differentiation with respect to an affine parameter and u is the impact parameter.

The null condition gives

$$\dot{r}^2 = \left(1 - \frac{b(r)}{r}\right) E^2 \left[e^{-2\Phi(r)} - \frac{u^2}{r^2} \right].$$

It is therefore useful to introduce the optical potential

$$U(r) = \frac{e^{2\Phi(r)}}{r^2}.$$

A circular null orbit located at a smooth interior radius $r > r_0$ must satisfy

$$\frac{dU}{dr} = 0,$$

which ...

Halo inference and adopted posterior inputs

The SPARC inference summarized in Table 2 and Fig. 1 selects two different halo descriptions for the subsequent Einstein–Cartan reconstruction. NGC3198 is described comparably well by DZ and Hernquist, with the lower-dimensional Hernquist model mildly preferred by the information criteria. For DDO154, DZ is strongly preferred over Hernquist in the baseline fixed-error comparison. Under the common intrinsic-scatter likelihood, DZ remains favored, although the strength of the preference depends ...

Conclusion

We have developed a SPARC-calibrated, posterior-propagated reconstruction of static galactic wormholes in Einstein–Cartan gravity. The Dekel–Zhao and Hernquist halo profiles were fitted to the rotation curves of NGC3198 and DDO154 using Bayesian posterior sampling. Hernquist is mildly favored for NGC3198, whereas DZ is preferred for DDO154, including when both halo families are compared using the same intrinsic-scatter likelihood. Joint posterior samples preserving the parameter covariance were ...

CRedit authorship contribution statement

Yohenba Soibam: Software, Investigation, Formal analysis, Conceptualization, Data curation, Methodology, Visualization, Writing – original draft, Writing – review &

editing. **M. Zeeshan Gul:** Writing – review & editing, Validation, Formal analysis, Methodology. **Kangujam Priyokumar Singh:** Validation, Supervision, Conceptualization, Writing – review & editing. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

References (63)

F.W. Hehl *et al.*

[Metric-affine gauge theory of gravity: field equations, noether identities, world spinors, and breaking of dilation invariance](#)

Phys. Rep. (1995)

N. Sarkar *et al.*

[Yukawa–Casimir wormholes within Einstein–Cartan gravity framework](#)

Phys. Dark Universe (2025)

A. Ashraf *et al.*

[Orbital motion and QPOs testing around rotating hairy black holes in Horndeski gravity](#)

Phys. Dark Universe (2025)

A. Ashraf *et al.*

[Exploring hu–Sawicki gravity through circular motion with trajectories and QPOs testing around a black hole](#)

Phys. Dark Universe (2025)

F. Rahaman *et al.*

[Possible existence of wormholes in the central regions of halos](#)

Ann. Phys. (2014)

M. Khatri *et al.*

[Existence of wormhole in Dekel–Zhao dark matter halo](#)

Phys. Dark Universe (2025)

A. Errehymy *et al.*

[Possible wormholes in generalized geometry–matter coupling gravity induced by the Dekel–Zhao dark matter profile](#)

Ann. Phys. (2025)

N. Sarkar *et al.*

[Testing energy conditions and light deflection in \$f\(Q\)\$ wormholes induced by Dekel–Zhao dark matter profile](#)

Phys. Dark Universe (2025)

M. Prajapati *et al.*

[A possible formation of galactic wormholes via the hernquist dark matter energy profile](#)

Phys. Dark Universe (2026)

A. Ashraf *et al.*

Thermal fluctuations, particle collision, QPOs, and emission energy around a Schwarzschild black hole immersed in Dehnen-type dark matter halo

Phys. Dark Universe (2025)



[View more references](#)

[View full text](#)

Recommended articles

No articles found.

© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

